

DOC327
Working With Data

Lecture 01 - Databases and Data Sets

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Lecture Summary

- Importance of Data in our life
- What is a dataset
- How can we combine a dataset
- What is a relational Database

Importance of Data

- Improve People's lives.
- Make informed decisions.
- Stop Molehills from turning into mountains.
- For Improving Processes.
- Data helps you understand performance.
- Find the best solution to a given problem.
- Know what you are doing well.
- Get the results you want.





iPhone 6s	VS	Galaxy S7
$808.95 as of Apr. 22nd	PRICE	$741.16 as of Apr. 21nd
750 x 1334	RESOLUTION	1440 x 2560
4.7 inches	SCREEN SIZE	5.5 inches
64 GB	STORAGE	32 GB
2 GB	RAM	4 GB
1715 mah	BATTERY	3600 mah
5.04 ounces	WEIGHT	5.54 ounces

Simple Comparison Making Inferences on Differences & Patterns

Beer and Diapers The Perfect Couple



- “One Midwest grocery chain used their data to analyze local buying patterns. They discovered that when men bought diapers on Thursdays and Saturdays, they also tended to buy beer. Further analysis showed that these shoppers typically grocery shop on Saturdays. On Thursdays, however, they only bought a few items. The retailer concluded that they purchased the beers to have it available till the upcoming weekend. The grocery chain could use this newly discovered information in various ways to increase revenue. For example, they could move the beer display closer to the diaper display and, they could make sure beer and diapers were sold at full price on Thursdays.”

Answer below questions?

- Who were the people who went buying the items in the case study?
- What were the major items they bought and why do you think?
- What is the discovery that the data analyst produced in the case study?
- How did it become a positive discovery for the organization?
- What is the decision the grocery chain made up on the finding?
- What are the other possible discoveries? (Suggest a few of your own.)

What is a Data Set?

Datum	Datum	Datum	Datum
Datum	Datum	Datum	Datum
Datum	Datum	Datum	Datum

- Data = many Datums
- Datum = 1 piece of info

- A record is a row of datums

- A data set is a structured arrangement of datums

  or

- A file that contains one or more **records** of data

Data Set

- Consider student details.
- A record belongs to an individual.
- Variables give meaning to the record and show the characteristics of individuals in the data set and the variation.

Age	Gender	Student ID
23	Male	2016058
18	Female	2017356
25	Male	2018987
22	Female	2015968



Variables

User ID	Address	Contact No
1	Jayawrdenapura	7714568
3	Wellawatta	4125888
2	Homagama	7605851
5	Kaduwela	7125645
6	Maharagama	7012625
8	Nugegoda	1155522
4	Bambalapitiya	7122411
7	Colpetty	5555824
9	Galle	7100000
10	Kandy	8122584
11	Anuradhapura	7822152
13	Jaffna	7200100
12	Matara	7400000
15	Ampara	7522145
14	Godagama	7811225

ID	Name	Week 01	Week 02	Week 03	Week 04
1	Saman	10000	20000	1000	5000
2	Kamal	8000	8000	8000	2000
3	Gihan	12000	11000	3000	8000
4	Ryan	12000	15000	18000	20000
5	Samali	50000	50000	20000	30000
6	Chamal	50000	30000	20000	20000
7	Gayani	1200	5000	6000	8000
8	Randima	12000	50000	55000	25000
9	Nethma	60000	65000	25000	20000
10	Piyath	5000	10000	11000	8000
11	Jagath	8000	12000	26000	30000
12	Waruna	500	14000	31000	32000
13	Nishan	25000	16000	25000	80000
14	Sandamal	2000	50000	56000	36000
15	Jananie	30000	2000	25000	55000

User IDs	Names
1	Saman
2	Kamal
3	Gihan
4	Ryan
5	Samali
6	Chamal
7	Gayani
8	Randima
9	Nethma
10	Piyath
11	Jagath
12	Waruna
13	Nishan
14	Sandamal
15	Jananie

Simple VLOOKUP Exercise

- Hint(Use the VLookUp example in the given sheet.)

How to combine a Data Set?

- Look for matching values and variables
- Use tools such as VLOOKUP, and Power Pivot to combine values of tables
- Link pieces of data through the values

What Is A Database?

- A database is a collection of data sets
- All data that is used in day-to-day lives and businesses is tracked and stored.
- The data that is stored need to be linked through relationships so that we can analyze and make decisions.

Different Types of Databases

- Centralized Database
- Distributed Database
- Relational Database
- Object-oriented Databases
- Hierarchical Databases
- Network Databases
- NoSQL Database

Relational Databases

- The most common type of databases.
- A database structured to recognize relationships among stored items of information.
- This type of database will have data sets that are linked through relationships stored within the data model.
- MySQL, Oracle, and SQL Server are all examples of relational database software.

Why Relational Databases?

- Organizations have access to huge amounts of data.
- This data continues to grow on a daily basis
- Saving data provides organizations, the ability to automate their tasks and improve decision making.
- Organizations use data to make decisions based on quantitative analysis instead of a gut feeling.

Why Relational Databases? [Continued]

- Often, related data is stored in different data sets
- For example, one data set may store a customer's name, phone number, and address. While another might store the transactions made by that same customer.
- Organizations need to combine different data sets to provide business insights.
- Requirement for data with high accuracy and ease of access.

Further Reading

- Beer and Diaper Case Study.
- Currently, Popular/Trending Database Types.
- Currently, Popular Relational Databases and Why?.

References and Further Reading:

- Connolly, T., Beggs, C. and Strachan, A., (2014). Database Systems. 5th Edition. London: Addison Wesley.
- Date, C. J., (2004). An Introduction to Database Systems. 8th Edition. India: Person Education
- Watt, A. and N. Eng. (2014). Database Design – 2nd Edition. Victoria, B.C.: BC campus.